



ANALYSTLAB AFRICA
Data Analytics Internship Programme
Official Weekly Assignment
Week 2: Business Analytics Case Study

Document Control

Programme	Data Analytics Internship
Week	Week 2
Project	Business Intelligence & Interactive Dashboard Development
Version	1.0
Issue Date	_____
Submission Deadline	End of Week 2 (Sunday, 11:59 PM WAT)

Business Scenario

You are a Junior Business Intelligence Analyst at AnalystLab Africa Consulting. Your team has been engaged by a national retail company to design an executive dashboard that enables management to monitor sales performance, profitability, customer behavior, and regional performance.

Senior management requires an interactive dashboard that transforms raw business data into meaningful insights, allowing them to make informed strategic decisions. Your responsibility is to prepare the data, build a professional dashboard using Microsoft Power BI, and provide actionable recommendations based on your analysis.

Learning Objectives

By the end of this assignment, you should be able to:

- Understand the role of Business Intelligence in decision-making.
- Import and transform datasets using Microsoft Power BI.
- Build interactive dashboards using appropriate visualizations.
- Create meaningful Key Performance Indicators (KPIs).
- Analyze business performance using interactive reports.
- Communicate business insights professionally.
- Recommend data-driven business strategies.

Dataset

Choose **ONE** of the following datasets.

Option 1

Superstore Sales Dataset

<https://www.kaggle.com/datasets/vivek468/superstore-dataset-final>

Option 2

Global Superstore Dataset

<https://www.kaggle.com/datasets/apoorvaappz/global-super-store-dataset>

Business Questions

Your dashboard should answer the following business questions:

1. What is the overall sales performance of the company?
2. Which regions generate the highest sales and profit?
3. Which customer segments contribute the most revenue?
4. Which product categories perform best?
5. Which products are the most profitable?
6. What trends can be observed over time?
7. What recommendations should management implement to improve business performance?

Assignment Tasks

Part 1 – Business Understanding

Research the importance of Business Intelligence and dashboards in supporting organisational decision-making.

Prepare a **1–2 page Business Intelligence Overview Report** explaining:

- What Business Intelligence is.
- Why organisations use dashboards.
- How dashboards improve decision-making.
- The business objective of your selected dataset.

Part 2 – Data Preparation

Using Microsoft Power BI:

- Import the dataset.
- Review the dataset structure.
- Check for missing values.
- Remove duplicate records where necessary.
- Correct inappropriate data types.
- Perform basic data cleaning and transformation using Power Query.

Part 3 – Dashboard Development

Create a professional Power BI dashboard containing:

KPI Cards

- Total Sales
- Total Profit
- Total Orders
- Average Sales
- Profit Margin

Visualizations

Include at least:

- 2 Bar Charts
- 2 Column Charts
- 1 Line Chart
- 1 Pie or Donut Chart
- 1 Map (if applicable)
- 1 Matrix or Table
- Slicers/Filters for interactive analysis

Ensure the dashboard is clean, organized, and suitable for executive decision-making.

Part 4 – Business Insights

Using your dashboard, provide:

- Five key business insights.
- Three business risks.
- Three business opportunities.

Support each insight using evidence from your dashboard.

Part 5 – Business Recommendations

Provide at least **five actionable recommendations** that management can implement based on your analysis.

Each recommendation should be supported by insights obtained from your dashboard.

Deliverables

- Business Intelligence Overview Report (PDF or Word)
- Microsoft Power BI Project File (.pbix)
- Dashboard Export (PDF or PNG)
- Executive Summary Report
- GitHub Repository
- Upload all files to Google Drive (Anyone with the link can view)
- **Submit using the official Week 2 Submission Form Assessment Criteria**

Criteria	Weight
Business Understanding	10%
Dataset Inspection	15%
Dashboard Design	25%
Data Visualisation	20%
Business Insights & Recommendations	15%
Professionalism & Documentation	15%

Professional Development Requirement

As part of building your professional portfolio:

- Update your GitHub repository with your Week 2 project.
- Publish one LinkedIn post summarizing:
 - The business problem.
 - Dashboard screenshots.
 - Key insights.
 - Skills developed.
- Tag **AnalystLab Africa** on LinkedIn.
- Publish one X (Twitter) post summarizing your project.
- Tag **@analystlabafri**c on X.
- Include the hashtag **#AnalystLabAfrica** on both LinkedIn and X.

Professional Branding Reminder

AnalystLab Africa encourages every intern to document their learning journey professionally on LinkedIn and X. Consistently sharing your work helps build your portfolio and increases your visibility to recruiters and employers.

Submission Checklist

- ☐ Business Intelligence Overview Report
- ☐ Power BI Project File (.pbix)
- ☐ Dashboard Export (PDF or PNG)
- ☐ Executive Summary Report
- ☐ GitHub Repository Updated
- ☐ LinkedIn Post Published
- ☐ X Post Published
- ☐ Google Drive Link Submitted

Minimum Dashboard Requirements

Your dashboard **must** include:

- At least **5 KPI Cards**
- At least **8 visualizations**
- At least **2 slicers or filters**
- A professional layout with appropriate titles
- Consistent colors and formatting
- An executive summary page or insights section